

SECTION-D

No. of Printed Pages : 4

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Roll No.

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain construction and working principle of Babcock and Wilcox recovery furnace with a neat diagram.

Q.24 Write short note on any two of the following

- a) Cascade type Evaporator
- b) Lime kiln
- c) black liquor oxidizers

Q.25 Calculate the water evaporated from B/L for following data. Solid concentration of weak B/L = 15%, Solid concentration of heavy B/L=51%, feed rate of weak B/L = 325 Kg/hr

6th Sem / Chemical (Pulp & Paper)

Subject : Chemical Recovery

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 The chemical formula for salt cake is _____

- a) CaOH b) Na_2SO_4
- c) $(\text{CaOH})_2$ d) None

Q.2 Concentration of dissolved solids is measured in

- a) mole% b) mole fraction
- c) percentange d) none

Q.3 Write equivalent weight of Soda ash is

- a) 30 b) 53
- c) 65 d) 90

Q.4 Expand BPR

- a) Blue point Rise b) Blow point Rise
- c) Boiling point Rise d) none of the above

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Q.5 Write molecular weight of lime Mud

- a) 35 b) 78
- c) 100 d) 38

Q.6 Concentration range of Weak black liquor is

- a) 45-55% b) 56-65%
- c) 14-18% d) None

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Expand MEE

Q.8 What is active alkali

Q.9 Write causticizing reaction

Q.10 What is Reduction efficiency.

Q.11 Mention any one physical characteristics of black liquor.

Q.12 Write the name of main constituent of green liquor

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Define total alkali and causticity.

Q.14 Draw a neat sketch of any one evaporator.

Q.15 Explain briefly Startup and shutdown procedure of MEE

Q.16 Convert 725 gram NaOH as such into NaOH as Na_2O .

Q.17 Explain the principle of cyclone type direct contact evaporator.

Q.18 Write a note on lime mud handling

Q.19 Write a short note on dreg washer from black liquor

Q.20 Write reactions in reduction zone in recovery furnace

Q.21 Write a short note on Physical properties of soda.

Q.22 Define heat capacity of black liquor.